

## PITCH DECK

ai writes the code.  
math approves the merge.

ZK-attested · spec-driven code review for the AI-synthesis era.



## AUTHOR

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## TRACKS

2 · Specification / 3 · Spec-driven dev

## LIVE

[veridict.zkred.tech](https://veridict.zkred.tech)

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## INTRODUCTION

**Most code review today rests on a single named maintainer.** AI synthesis makes that worse. Review volume now exceeds what any one human can verify by reading. Veridict is a merge gate that decouples **qualified-reviewer-ness** from **identity**. Three formal proofs verify the PR. One ZK signature authorises it. Zero identities revealed.

## 01 / INTRODUCTION

# hello, here is veridict.

**A merge gate for synthesized code.** Three formal proofs verify the PR (mypy, pytest, Z3). One Longfellow ZK signature authorizes it. Zero identities revealed.

## 02 / THE PROBLEM

# open source is leaking trust.

Four incidents. One pattern. Different projects, same architectural hole.

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INCIDENT 01

Dec 2024

## bincode

Maintainer rewrote git history while migrating off GitHub. The questions became doxxing. The project shut down permanently.

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INCIDENT 03

Jan 2024

## curl

“My job is increasingly fact-checking the AI,” wrote Daniel Stenberg. Maintainers drown in plausible AI PRs that fail invariants.

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INCIDENT 02

Mar 2024

## xz-utils

“Jia Tan” earned maintainer status over two years, then shipped a backdoor that nearly compromised sshd on every major Linux distro.

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INCIDENT 04

Feb 2025

## vibe-coding

Karpathy coined it. The practice went mainstream anyway: ship when it looks right. Verification skipped at scale.

## 03 / THE THESIS

# the thesis.

## DECOUPLE

**qualified-reviewer-ness** from **identity**.

The architectural hole in open-source review is not the absence of clever tooling. It is the lack of separation between being a qualified reviewer and being a known identity.

## REPLACE WITH

two separable primitives.

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01

Formal gate

mypy + pytest + Z3

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02

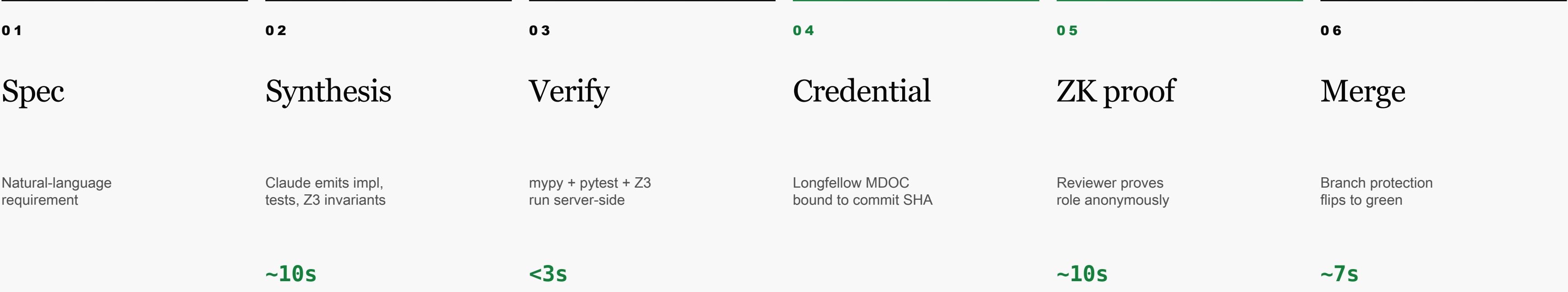
ZK credentials

Longfellow MDOC

04 / HOW IT WORKS

six stages.  
one signature.

End-to-end on Cloud Run. The reviewer clicks once.



End-to-end ~30 seconds from PR comment to merge-eligible commit status.

# what was built.

Six concrete pieces shipped in the timebox. Every one of them is a real file in the repo.

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## 01 Spec-check gate

mypy + pytest + Z3 results gate ES256-signed MDOC minting.

[issuer/server.py](#)

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## 03 Longfellow patch

Reviewer namespace org.example.reviewer registered.

[patches/add-reviewer-namespace.patch](#)

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## 05 GitHub branch gate

Bot pushes zk-review-gate commit status; merge gated at API level.

[.github/workflows/](#)

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## 02 Structured synthesis

Claude emits impl, pytest suite, and Z3 invariants in one call.

[issuer/synthesizer.py](#)

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## 04 Anti-replay MDOC

Transcript = sha256(owner/repo/pr/sha) bound inside ZK session.

[issuer/mdoc\\_builder.py](#)

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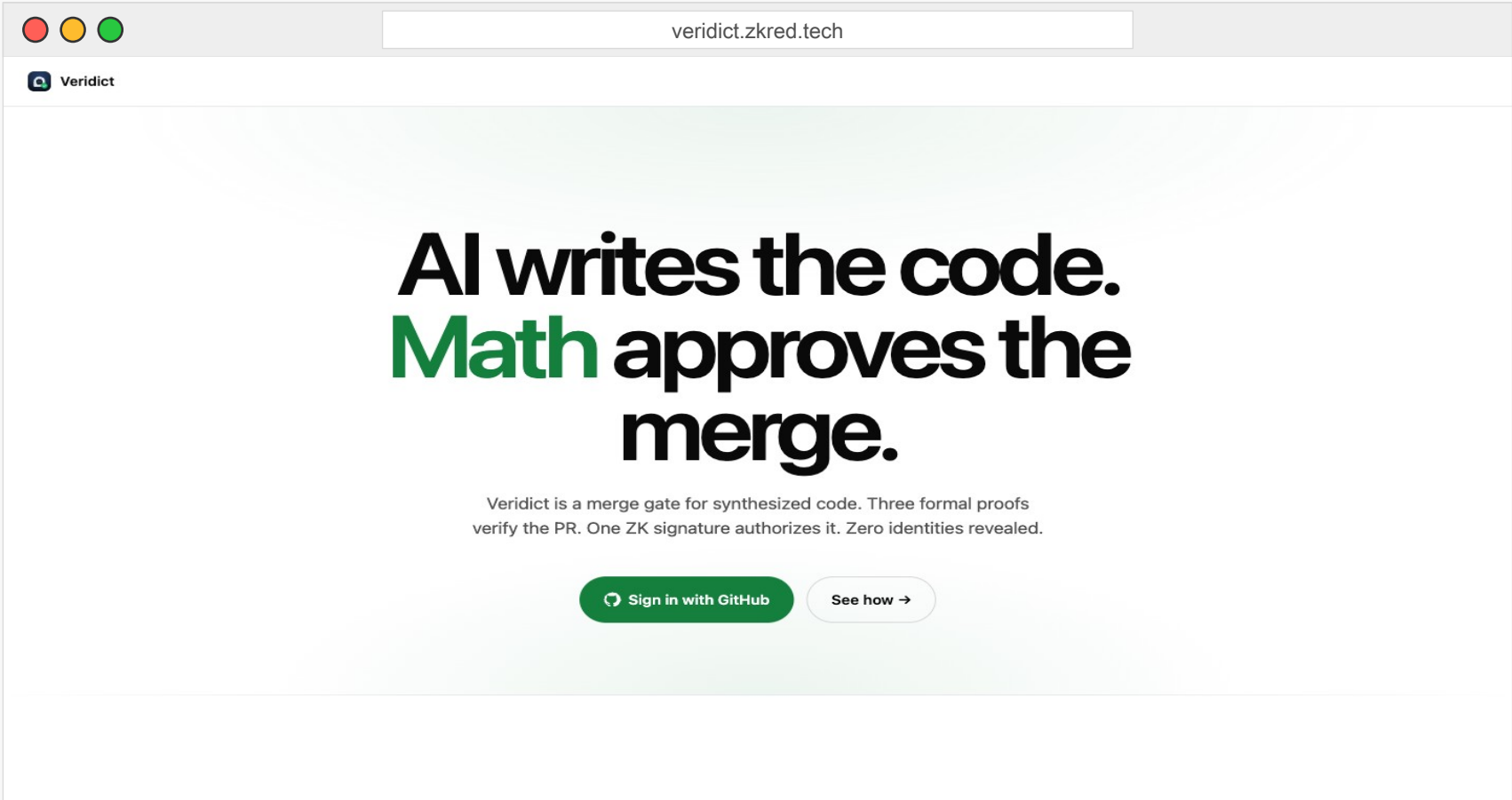
## 06 Pseudonyms + chips

HMAC-derived per-PR tags; identicons; role/age/repo-count buckets.

[backend/main.py](#)

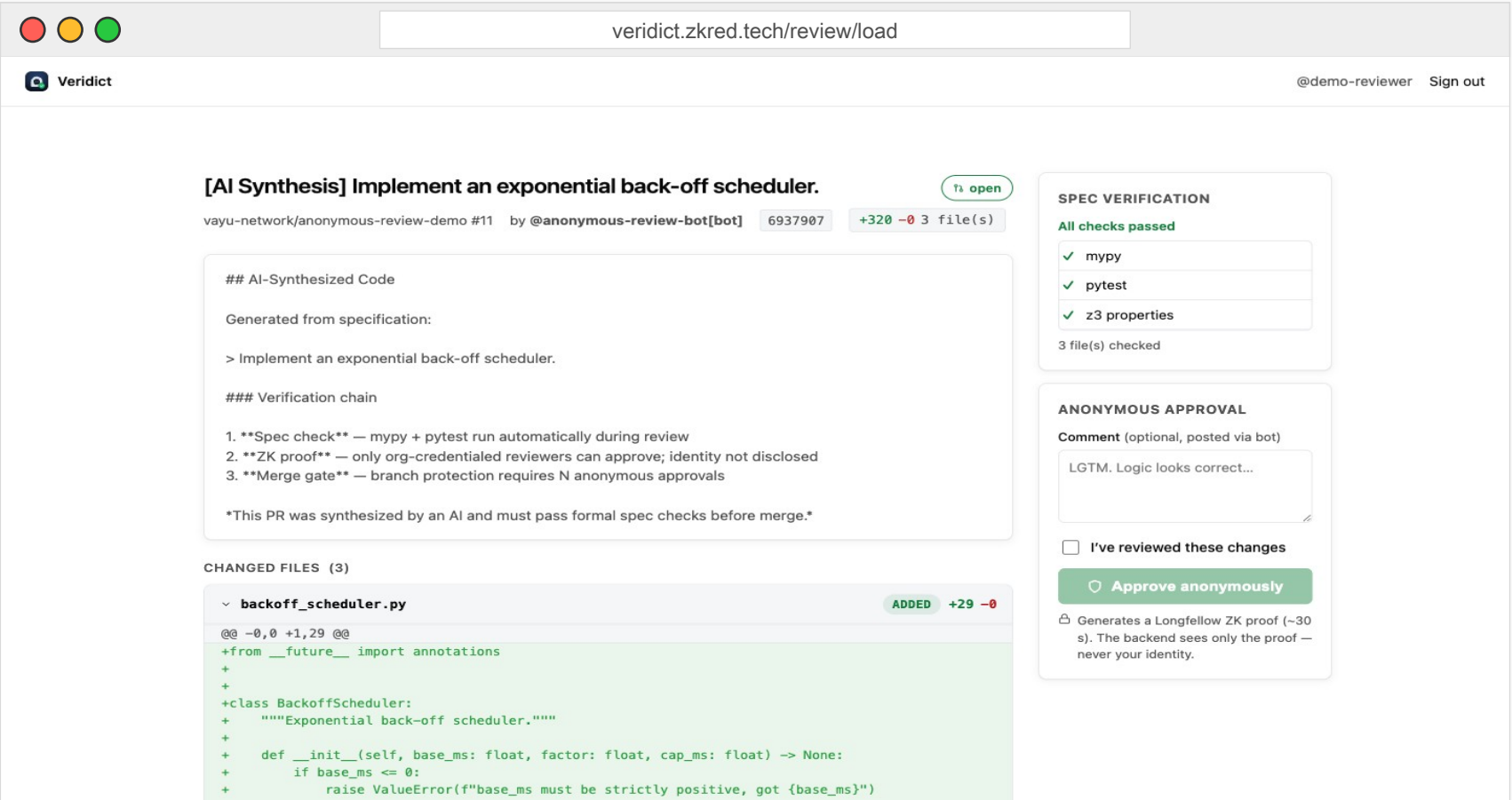
# it is real. it is deployed.

Two views from the live deployment: the landing page and the reviewer's PR view.



## LANDING

Sign in with GitHub · synth or review.



## REVIEWER VIEW

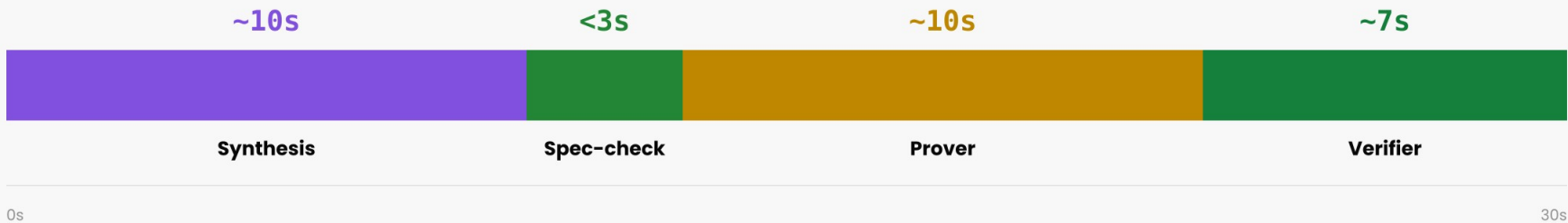
Spec diff, spec-check, approve anonymously.

measured on cloud run.

END-TO-END

~30 seconds

from PR comment to merge-eligible commit status.



PROOF SIZE

360 KB

VERIFIER RSS

365 MB

ANON APPROVALS

2 / 2

● LIVE NOW    Zkred/veridict

Cloud Run · 4 GB · branch-protection enforced

PIPELINE TIMINGS

|            |                    |      |
|------------|--------------------|------|
| Synthesis  | Claude → PR opened | ~10s |
| Spec-check | mypy + pytest + Z3 | <3s  |
| Prover     | Longfellow ZK      | ~10s |
| Verifier   | Longfellow ZK      | ~7s  |

ARTIFACTS

|              |                       |        |
|--------------|-----------------------|--------|
| Proof size   | per approval          | 360 KB |
| Verifier RSS | peak memory           | 365 MB |
| Credential   | MDOC, 10-min validity | ES256  |

DEPLOYMENT

|             |                        |          |
|-------------|------------------------|----------|
| Approvals   | anonymous + verified   | 2 / 2    |
| Branch gate | zk-review-gate context | enforced |
| Repo        | Zkred/veridict         |          |

# who sees what.

Anonymity is structural. Below: who can see which fact, from least private to most public.

|                             | REVIEWER | ISSUER | BACKEND / CI | PR AUTHOR + GH |
|-----------------------------|----------|--------|--------------|----------------|
| Reviewer's true identity    | • self   | ✓      | .            | .              |
| OAuth identity              | • own    | ✓      | .            | .              |
| Spec-check result           | • own PR | ✓      | ✓            | ✓              |
| ZK proof                    | • own    | .      | ✓            | .              |
| Pseudonym + reputation      | • own    | ✓      | ✓            | ✓              |
| Approval count (N / req'd)  | .        | ✓      | ✓            | ✓              |
| Other reviewers' identities | .        | .      | .            | .              |

✓ sees this · • own only (own credential, own PR scope) · · hidden

**Note:** anonymity is toward the verifier: PR author, CI, audit trail. **Not toward the issuer. Said out loud.**

## 09 / WHAT THIS ISN'T

# limits, said out loud.

A weekend prototype is allowed limits as long as it names them. Here are ours.

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## ANONYMITY

### No nullifier yet

Two browser sessions = two pseudonyms for one reviewer. Fix: derive a blinded ID inside the ZK circuit.

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## SPEC COVERAGE

### Python only, for now

mypy + pytest + Z3 only run on .py files. Non-Python PRs pass through unverified. Dafny, Rust, Lean are obvious next steps.

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## KEY CUSTODY

### Issuer mints the device key

For demo speed. A real deployment generates the device key in the browser so the issuer never sees it.

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## COMMENTS

### Single bot account

Every anonymous approval posts as @anonymous-review-bot. Pseudonyms + identicons inside the comment do the distinguishing.



# thank you.

*anonymity should be cheaper than identity.*

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TRY IT

**veridict.zkred.tech**

CODE

[github.com/Zkred/veridict](https://github.com/Zkred/veridict)